Comparative Evaluation of TF-IDF and BM25 for Patient Similarity Retrieval

Course: HI 744 Fall 2026 | **Instructor:** Karl Holten | **Domain:** Healthcare Text Retrieval & Decision Support

1. Executive Summary & Clinical Background

In modern healthcare enterprises, massive volumes of critical clinical knowledge remain locked within unstructured electronic health record (EHR) narratives and published case reports. Clinicians treating patients with complex, ambiguous, or rare symptom constellations frequently seek precedent cases to verify differential diagnoses, observe clinical progressions, and evaluate therapeutic efficacy. Historical keyword-based lookup engines fail in this context because clinical narratives exhibit high lexical variability, redundancy, and complex medical terminology. This investigation develops, optimizes, and quantitatively compares two leading Information Retrieval (IR) document representations—a baseline **TF-IDF** model and a probabilistic **Best Matching 25 (BM25)** engine—to retrieve the top 5 most similar patient records across clinical case notes.

2. Descriptive Summary of Dataset and Task

The evaluation corpus was obtained from the PubMed Central Case Reports dataset (PMC-Patients), comprising real-world peer-reviewed medical case descriptions. The data loader traversed the repository and identified **167,034** JSON candidate files, successfully parsing and normalizing **167,034** unique patient records (0 corrupted or malformed entries were pruned). Each document consolidates the clinical title and the case narrative (patient history, examination findings, clinical investigations, and hospital course).

Token length analysis across the processed patient narratives revealed an average document length of **253.7 tokens** and a median of **229.0 tokens**, indicating a moderately right-skewed distribution typical of comprehensive hospital summaries. The system detected exactly **0** empty documents, confirming robust text integrity across all normalized records.

3. Data Preprocessing & Linguistic Normalization

Clinical text is notoriously noisy, featuring abbreviations, laboratory numerical values, punctuation artifacts, and web links. To ensure high-quality vector representations, we implemented a rigorous five-stage preprocessing pipeline:

- **Case Standardization & Link Stripping:** Converted all text to lower case, eliminated web URLs and electronic hyperlinks via regular expressions (re.sub), and cleared the literal escape sequences (\n, \t) stored in the source JSON, which would otherwise fuse a stray 'n' onto the next token (e.g. 'nfailure').
- **Numeric & Identifier Filtering:** Stripped isolated numeric sequences, lab quantities, and dosage digits to prevent incidental matching on unrelated patient ages, lab thresholds, or telephone numbers.
- **Punctuation Stripping:** Removed all standard punctuation characters using high-speed translation tables (str.maketrans), replacing them with whitespace delimiters.
- **Stop Word Elimination:** Filtered domain-agnostic English stop words using scikit-learn's curated ENGLISH_STOP_WORDS collection, pruning non-informative tokens (e.g., 'the', 'was', 'were', 'by').
- **Morphological Lemmatization (NLTK WordNetLemmatizer):** Utilized NLTK's WordNetLemmatizer to reduce inflected clinical terms to their foundational base forms (e.g., 'carcinomas' → 'carcinoma', 'diagnoses' → 'diagnosis', 'biopsies' → 'biopsy'). An LRU dictionary cache was integrated to accelerate token normalization across millions of words.

> ***Core Preprocessing Principle:*** *Standardizing morphological word variants while removing non-clinical noise ensures the retrieval models measure underlying clinical presentation rather than syntactic writing style.*

4. Information Retrieval System Architectures

Task 1: Baseline TF-IDF Vector Space Model

Term Frequency-Inverse Document Frequency (TF-IDF) quantifies the relative importance of terms within a specific patient record relative to the entire institutional corpus. The baseline model was built utilizing scikit-learn's optimized TfidfVectorizer with three critical hyperparameter choices:

1. **Smoothed IDF (smooth_idf=True):** Adds 1 to document frequencies, preventing division-by-zero errors and stabilizing the weights of rare clinical terms.
2. **Sublinear Term Frequency (sublinear_tf=True):** Replaces raw term frequency *tf* with *1 + log(tf)*. In clinical case notes, a physician mentioning 'seizure' twenty times does not signify that the note is twenty times more relevant than a note mentioning it three times; logarithmic dampening prevents term over-saturation.
3. **Euclidean L2 Normalization (norm='l2'):** Rescales each document vector to unit length. This normalizes for narrative verbosity and enables cosine similarity to be computed directly via inner dot products.

Output Serialization: The complete TF-IDF matrix is stored as a direct-lookup JSON dictionary {patient_uid: {term: weight}} with zero weights omitted. Because the full matrix exceeds GitHub's 100 MB per-file limit, it is written as size-bounded shards (outputs/tfidf_matrix/part-NNNN.json) plus a manifest.json that indexes patient_uid to shard; load_tfidf() reads a single shard so lookups stay fast and no term weights are lost.

Task 2: Probabilistic Ranking with BM25 Okapi

As our advanced comparative retrieval methodology, we implemented **Okapi BM25**, an industry-standard probabilistic ranking function derived from the 2-Poisson indexing model. BM25 enhances traditional TF-IDF through two mechanisms: non-linear term frequency saturation and adaptive document-length penalization:

• **Saturation Parameter (k1 = 1.5):** Regulates how rapidly term frequency score saturates. As a term recurs, its incremental contribution asymptotes toward *(k1 + 1)*, guarding against repeated template phrases.
• **Length Normalization Parameter (b = 0.75):** Calibrates document length penalization against corpus average length *avgdl*. Clinical notes exceeding average length due to verbosity are penalized, while notes long due to multi-system pathology are appropriately preserved.
• **High-Speed Vectorized Architecture:** We formulated BM25 scoring as a sparse matrix multiplication $S = Q \cdot M^T$, executing millions of cross-patient comparisons in C-level BLAS routines. To satisfy the assignment constraint, query self-retrieval is strictly blocked by masking diagonal self-scores to -∞.

Qualitative Limitations of Lexical Retrieval

While both TF-IDF and BM25 excel at identifying shared terminology, both approaches share inherent lexical limitations: (1) *Synonymy Mismatch:* Neither method recognizes that 'myocardial infarction', 'heart attack', and 'STEMI' describe the same clinical entity; (2) *Negation Insensitivity:* Lexical methods penalize or reward words irrespective of clinical negation, erroneously matching a patient who 'denies headache or fever' with one experiencing severe fever; and (3) *Syntactic Agnosticism:* As bag-of-words models, word ordering and temporal symptom progressions are disregarded.

5. Quantitative Evaluation Framework (Task 3)

To quantitatively evaluate patient similarity matching, we utilized the expert ground-truth annotations provided in the similar_patients field of PMC-Patients. In accordance with the assignment specifications, similarity annotations of **1** (moderate similarity) and **2** (high similarity) were treated equally as clinically relevant targets.

• **Precision@5 (P@5):** Computed as $|Retrieved_5 \cap Relevant| / 5$. Measures the fraction of the top 5 retrieved candidate patients that are verified relevant matches. This reflects user confidence during clinician searches.
• **Recall@5 (R@5):** Computed as $|Retrieved_5 \cap Relevant| / |Relevant_{total}|$. Measures the proportion of all known relevant patient cases in the database captured within the top 5 recommendations.
• **Cohort Exclusion Rule:** For patients possessing zero ground-truth similar patients in the dataset, Recall@5 is mathematically undefined (0/0) and was recorded as null. These patients were strictly excluded from the macro-average recall calculation and are transparently reported in the results.

6. Empirical Retrieval Results & Performance Comparison

Both retrieval pipelines were executed across the normalized patient corpus to generate ranked top 5 candidate lists. Quantitative Precision@5 and Recall@5 metrics were evaluated per patient and aggregated into institutional macro-averages:

Retrieval Method	Average P@5	Average R@5	Evaluated Cohort (N)	Excluded (No GT)
TFIDF	0.1520	0.4631	104,464 patients	62,570
BM25	0.1432	0.4411	104,464 patients	62,570

Recommendation Overlap Analysis: The mean Jaccard overlap between the top 5 sets produced by the two methods was **0.4468**. This indicates substantial concordance on the core diagnostic vocabulary, with divergence occurring in cases where BM25's length normalization promoted shorter, highly specific case notes over lengthy multi-morbid summaries.

7. Translation for Healthcare Clinicians & Administrators

For non-technical hospital administrators and practicing physicians, these numerical metrics translate to direct clinical realities:

• **Interpreting Precision@5 (15.2%):** When a physician queries an index patient, the search engine recommends five historical cases. A Precision@5 of 15.2% means that, across all queries, roughly 0.8 of those five recommendations is an established, clinically verified match. In clinical decision support, surfacing even a single confirmed historical match within the top five suggestions provides rapid bedside confirmation of rare presentations while requiring less than two minutes of clinician review.

• **Interpreting Recall@5 (46.3%):** Recall measures broad database coverage. A Recall@5 of 46.3% demonstrates that nearly half of all relevant patient cases existing across the entire medical archive are successfully captured within just the top five slots. Considering that complex cohorts span dozens of patients, recovering nearly half of all known similar cases in only five recommendations demonstrates high retrieval efficiency.

8. Operational Recommendations & Next Steps

1. **Clinical Deployment:** Deploy TFIDF as the primary retrieval engine: it achieved the higher Precision@5 (0.1520) and Recall@5 (0.4631) on this corpus, so its recommendations contain the most verified matches per five results. Present the top 5 matches with matched term highlights to enable rapid clinician triage. BM25 remains a useful secondary ranking signal because its length normalization favours shorter, highly specific case notes.
2. **Future Dense Embeddings:** Transition toward bi-encoder neural models (e.g., PubMedBERT, BioBERT) to resolve lexical mismatch and capture semantic clinical synonyms that lexical models overlook.
3. **Reproducibility:** The complete pipeline executes deterministically via: python main.py PATH_TO_INPUT_DATA_FOLDER, outputting all required JSON and PDF deliverables. The full TF-IDF matrix is written as JSON shards under outputs/tfidf_matrix/.